



ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

College of Engineering

Department of Electrical and computer Engineering

Final Year Project on

“Title”

By:

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Chapter One

Introduction

(Chapter titles should stand as: Headlining 1, Times New Roman, size 14, centered, bold, and layout (24 before and 12 after the title))

The Introduction guides you to understand the significance of your research. It should start with a general overview of your subject prior to focusing on your main research question or hypothesis. *The basic format for paragraphs should be in:*

- *Times New Roman (Normal format), size 12, spacing 1.5 and its layout should be 6 before and after the paragraph.*

1.1 Background of The Study (Background information):

The subtitles should stand as: headlining 2, Times New Roman, size 12, centered, bold, and layout (6 before and after the title)

Strong research/project backgrounds sets the stage for your study and entice readers to read the rest of your writing. In this section, the author often provides a succinct overview of the historical developments in the literature that led to the current research issue. In case the research is multidisciplinary, it must specify the connections between many fields of study as well as the facets of each field that will be examined. Author should summarize the key advancements in their field of study and point out the primary gaps that still require filling. Stated otherwise, this section needs to provide a synopsis of your research. The section ought to be set up as follows:

- What is known about the general topic?
- What are the gaps or missing links that require attention?
- Why is it important to fill in those gaps?
- What are the goals and assumptions of your research?

Therefore, the background section should highlight the primary goals of the study and offer broad information on the subject of your research.

Please make sure you just talk about the most important and pertinent parts of the research that lead to your objectives.

(NB: Don't forget citing the source of information's while presenting the supportive idea of your topic, new findings and current state of art in the subject area)

1.2 Statement of the problem

The problem or issue that the research project seeks to answer should be precisely defined in the statement of the problem. It ought to be both manageable and broad enough for other experts in the field to find it interesting. It should clearly present:

- The problem that you are motivated to mitigate. It should include both domain problem and scientific problems.
- Domain problem means the existing problem which you want to deal with.
- Scientific problem means the limitation of existing techniques in terms of solving the named problem. Existing techniques' strong side and limitation in mitigating similar problem should be clearly summarized in statement of the problem. Thus, you need to present the details of existing problem strong side and limitation in Literature section with proper citations/references, but not here. Here, you need present summary of limitation of existing problem in mitigating the project domain problems
- Then, your proposed method/technic in terms of its advantage/width/coverage/ importance in filling the limitation or gaps of existing methods should be presented here.
- Thus, your proposed method/technique should have contribution to the existing scientific methods. The state of art/innovativeness of your proposed technique /method should be clearly presented

1.3 Objective of the study

1.3.1 General objective of the study

A general objective is a single sentence that sums up the entire purpose of a study or research endeavor.

1.3.2 Specific Objective of the project

The more manageable, achievable targets that contribute to achieving the main objective of research project are outlined in the specific objectives. These are always sharper in focus, with each new goal building on the previous to produce a well-defined course of actions. Research project scopes can be defined based on clearly defined and presented specific objectives. each specific objective should be SMART (specified, measurable, attainable, relevant, and time-bound). Present your specific objectives in bulleting format using the identifiers word like

- To develop
- To model
- To optimize

NB: the number of specific objective should be equal to the identified problems in statement of the problem, means it should be drawn from statement of the problem. In addition, all the specific objectives should have clearly developed methodology and results

1.4 Scope of the study

The scope of a study outlines the parameters that the investigation will operate under and explains the extent to which the research area will be studied in the study. It should present the boundary of your research project work in a single paragraph.

1.5 significance of the study

1.6 Outlines of the research project

Chapter Two

Literature review

2.1 Theoretical Review of the literature

A description of what has already known about this topic and short discussion of why the background studies are not sufficient should be presented here. List the current state of knowledge or basic science in the field that are crucial for the proposed study. Give a synopsis of the fundamental background knowledge you have learned about the subject from your literature review (you can add details from the class and book, but outside sources should comprise the majority of the information) using proper citation or mentioning the source of information using standard referencing format (See IEEE standard referencing format).

2.2 Empirical review

This subsection should describes/ presents the technical review of existing recent literatures (published within the last 3-5 years), which are published in international acceptable and repeatable journals. Describe a number of important studies that have already been conducted in your proposed research objective (using IEEE style citations). *Describe the shortcomings and strong sides of the existing studies.* Put differently, which unanswered question(s) do they leave open that you would like to investigate (for each specific objective)?. Pick the problems/inquiries that you would like to look into further. It is advised that you give the empirical review for each specific objective independently. This will help you to formulate the research gap that connects each gap to the corresponding specific objective.

2.2.1 Empirical review based on specific objective one

2.2.2 Empirical review based on specific objective two

2.2.3 Empirical review based on specific objective three

2.3 Research Gaps

Based on the empirical review provided in subsection 2.2, develop the research gaps. The limitation of current methodologies serves as the foundation for the development of research gaps. As a result, you should either offer a fresh, creative idea or add to the body of information already in existence to address a need. Keep in mind that the number of research questions (particular objectives) and the generated research gaps should match. You should also have a

well-defined approach and methodology to handle each gap, and your research findings should result in outcomes that are clearly presented and supported by scientific evidence

Chapter Three

Method and Design

This section will include a discussion of your suggested methods for obtaining information and evaluating the ideas you are researching, as well as the procedures for creating models or source codes and creative solutions. You are welcome to attempt coming up with an original or creative solution, but it is not required. Look for methodologies in journal articles that are frequently used to assess current approaches, methods, and language proficiency in the area you have chosen, then adapt them to fit your needs. Describe the following points (provide subsections).

- How and why would you go about gathering the data if any?
- Explain the overall approach you decide to take in order to test your hypothesis or hypotheses.
- Justify this approach's superiority for your needs over others.
- Those in attendance: Who and why would you test?
- Explain the model/sample you would test and the rationale for your selection.
- Provide details on uniqueness of your approach towards the importance of existing engineering design/problem solving, and socioeconomic advantages

Finally, using a mathematical model (if any), simple procedures, and a flowchart to simplify the presentation, build a process that guided you to accomplish each specified goal independently. Write about theoretical studies, experimental set up required, and method of analysis

Chapter Four

Results and Discussions

Give the general introduction of the chapter based on the overall results that you are going to present in this chapter. Present the anticipated findings from your study. Remember to align the project's aims and objectives with the results that are developed. Present your results and discussion based on the following formats.

4.1 **results for specific objective one**. NB: replace the highlighted red sentence by your specific objective one. Example: 4.1 Modeling/optimization/designing of.....

Then, discuss your results in detail

4.2. Results and discussion on specific objective two. Follow the same procedure

4.3.....

5. Conclusion and recommendations

5.1 Conclusion

5.2 Recommendation

References

In this section, Please present/lists the references you have used while formulating the research problems, developing your methodology and validating the results. Note down below the references consulted for the proposal. List the references according to the citation used, with the first cited reference at the top as [1] and the last at the bottom as [n]. For references, use the IEEE standard format. NB: only cites the references you have used in compiling your work, but not include others

